

Chapter 3. GROUPS

3.2 Subgroups



Today we will

- Learn what a **SUBGROUP** is.
- See a few examples.
- Learn a way to check whether a given subset of a group is a subgroup.

What is a subgroup?

Definition 3.13. A subset H of a group G is called a **subgroup of G** if H is itself a group using the operation of G .

Examples 3.14.

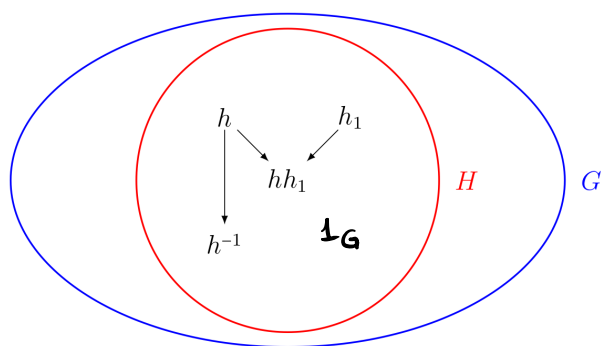
1. Each of the **additive groups** $\mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$ is a subgroup of the larger ones.
2. The subset of even permutations A_n is a subgroup of S_n , called the **alternating group of degree n** .

How to find out whether a subset is a subgroup

SUBGROUP TEST. A subset H of a group G is a subgroup of G if and only if the following three conditions are satisfied.

- (1) $1_G \in H$, where 1_G is the identity element of G .
- (2) If $h \in H$ and $h_1 \in H$, then $hh_1 \in H$.
- (3) If $h \in H$, then $h^{-1} \in H$. Here h^{-1} denotes the inverse of h in G .

In this case, H has the same unity as G , and if $h \in H$, its inverse in H is the same as its inverse in G .



Proof. $\Rightarrow$ Trivial

$\Leftarrow$ Suppose that H is a subgroup of G ,
and let us check that H satisfies ①, ② and ③,

(1) Write $e \in H$ to denote the identity
element of H . Then

$$e^2 = e \cdot e = e,$$

and by the Cancellation law in G
we obtain that $e = 1_G$. So, $1_G \in H$
and (1) follows.

(2) follows because H is closed under the
operation of G , since we are assuming

that H is a subgroup of G .

(3) For $h \in H$, suppose that $h' \in H$ is the inverse of h in H . Write h^{-1} to denote the inverse of h in G .

Goal: $h^{-1} \stackrel{?}{=} h'$.

Since h' is the inverse of h in H , we have that

$$hh' = 1_G = h'h$$

But $hh' = 1_G = hh^{-1},$

and by the Cancellation Laws we obtain that $h^{-1} = h' \in H$



Example 3.15. Let $n \geq 0$ be an integer. Then the set of all multiples of n , namely, $n\mathbb{Z} = \{nk \mid k \in \mathbb{Z}\}$ is a **subgroup of the additive group** $(\mathbb{Z}, +)$.

Using the Subgroup Test we need to check that $n\mathbb{Z}$ satisfies (1), (2) and (3).

$$(1) \quad 0 \stackrel{?}{\in} n\mathbb{Z}$$

$$0 = n \cdot 0 \in n\mathbb{Z} \quad \checkmark$$

$$(2) \quad a, b \in n\mathbb{Z} \stackrel{?}{\Rightarrow} a + b \in n\mathbb{Z}$$

$$a, b \in n\mathbb{Z} \Rightarrow a = nk_1, b = nk_2, \text{ for some } k_1, k_2 \in \mathbb{Z}$$

$$\text{Then } a + b = nk_1 + nk_2 = n(k_1 + k_2) \in n\mathbb{Z}$$

$$(3) \quad a \in n\mathbb{Z} \stackrel{?}{\Rightarrow} -a \in n\mathbb{Z}$$

$$a \in n\mathbb{Z} \Rightarrow a = nk_1, \text{ for some } k_1 \in \mathbb{Z}.$$

$$\underline{-a = -nk_1 = n(-k_1) \in n\mathbb{Z}}$$

